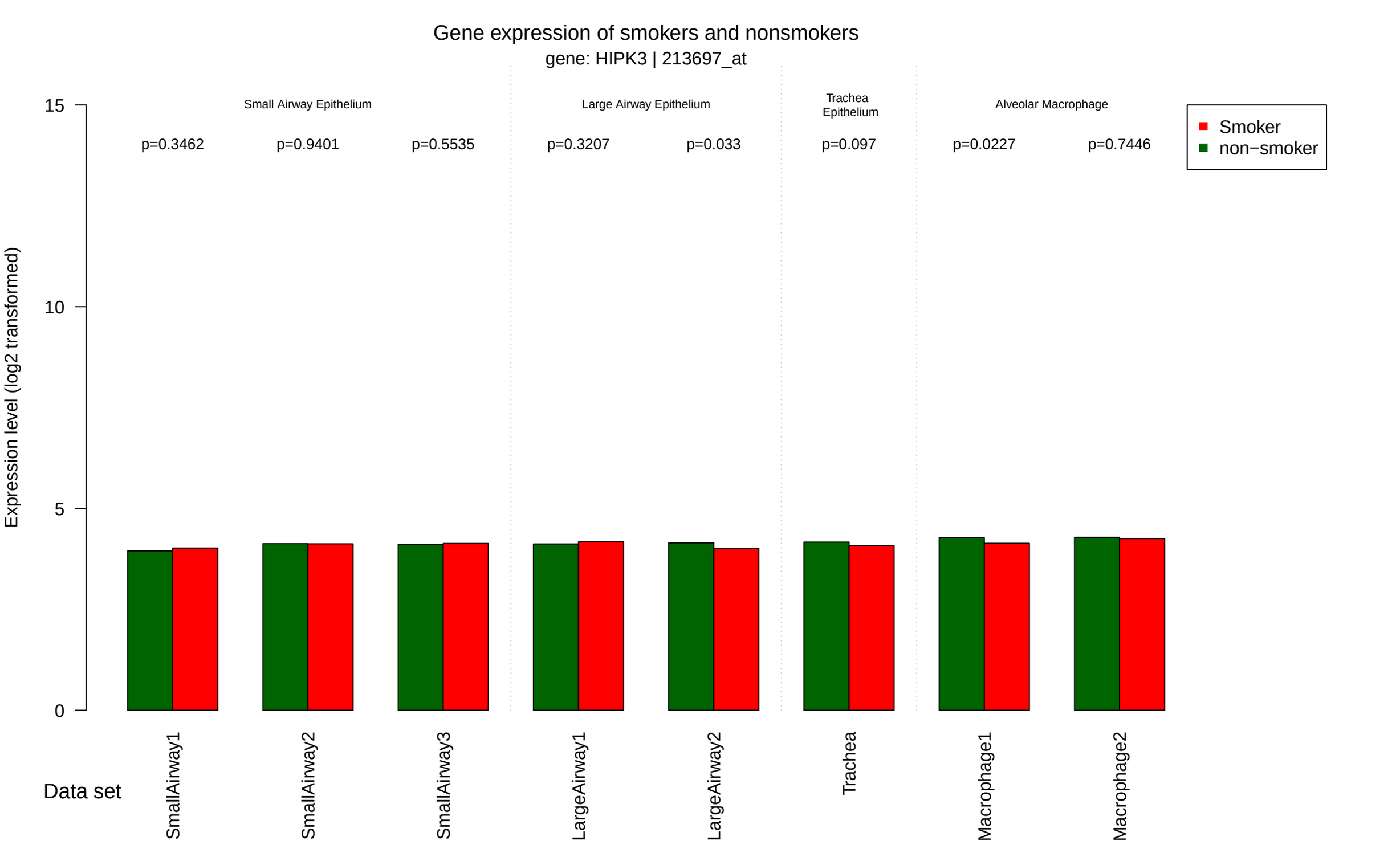
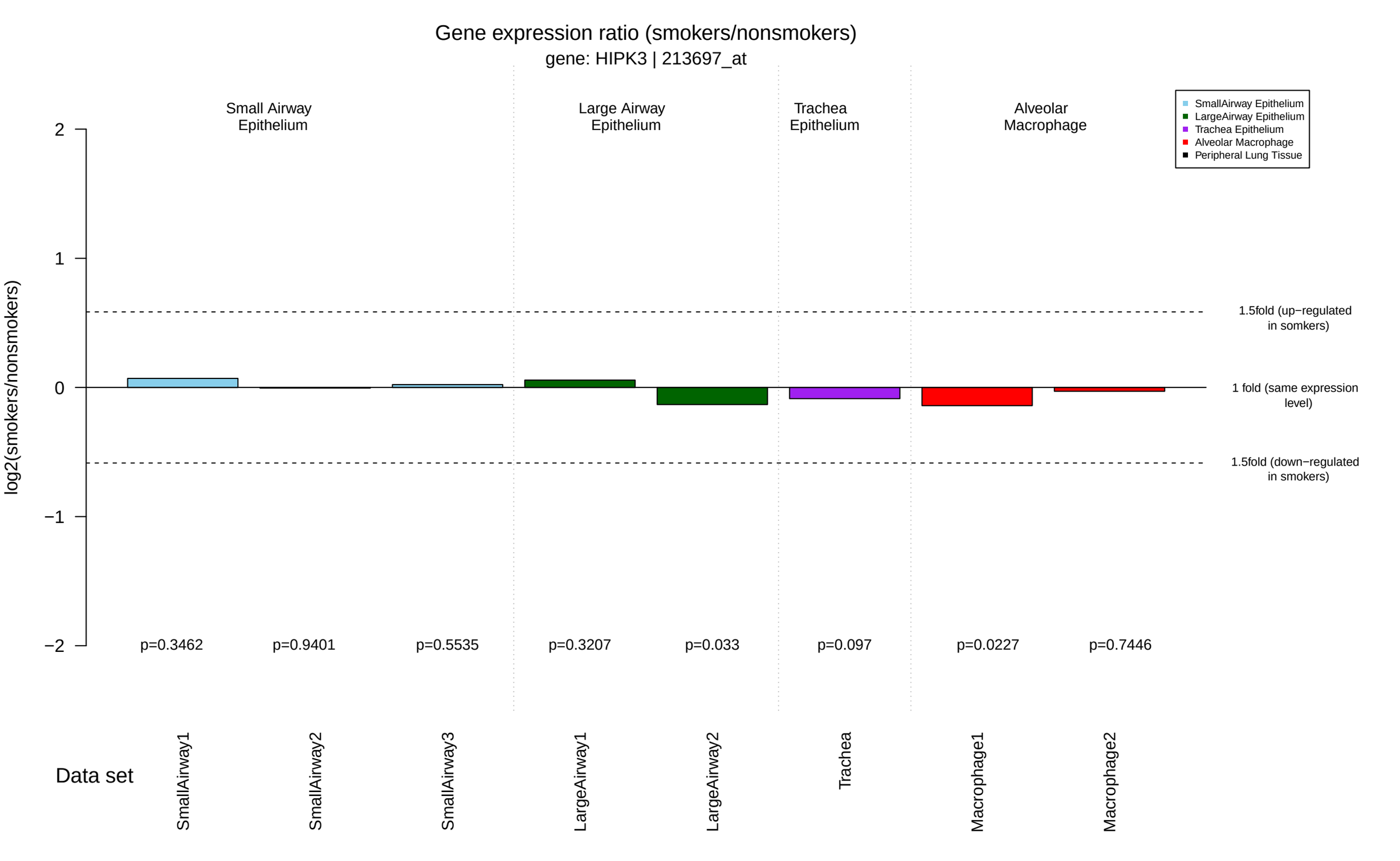
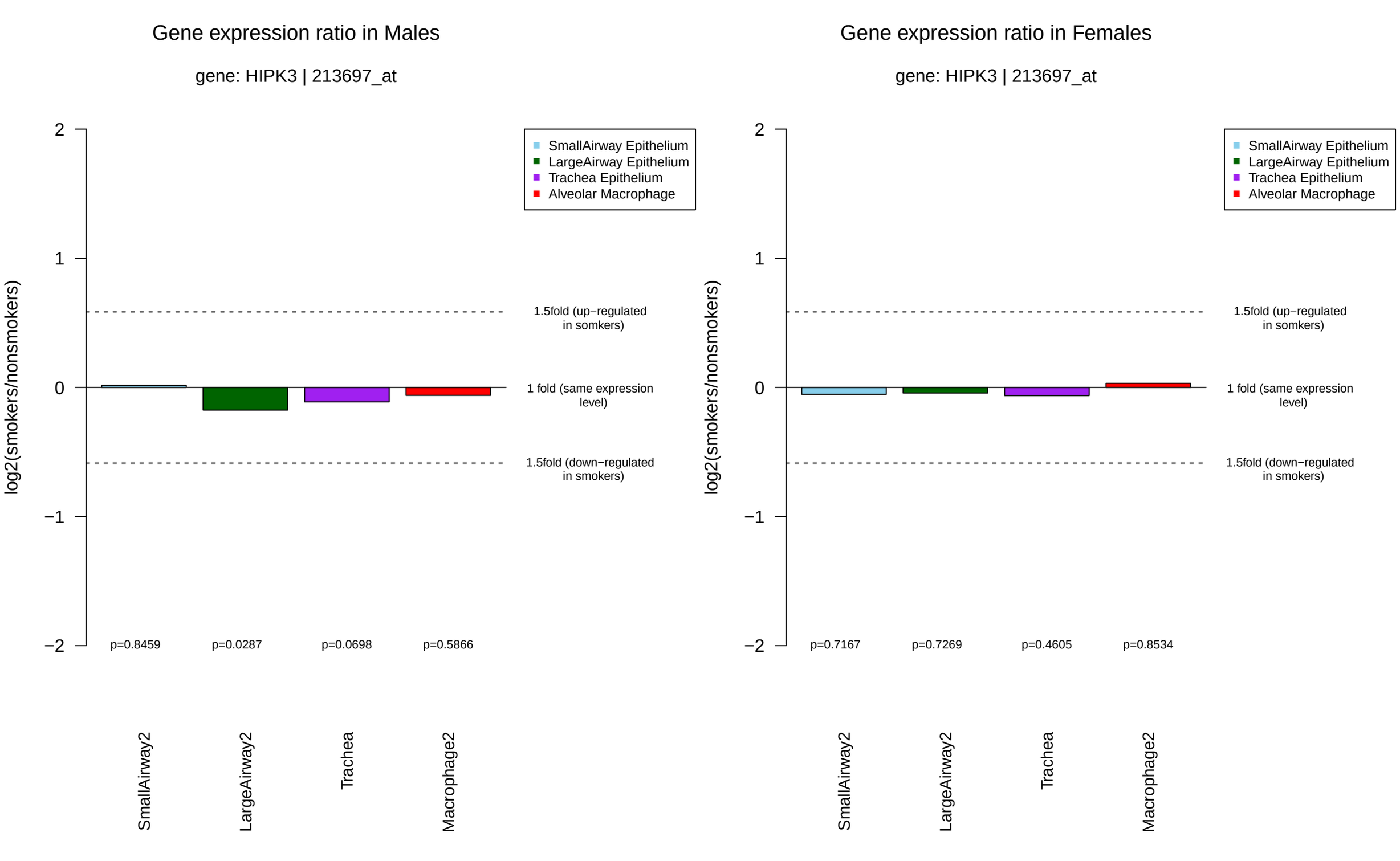
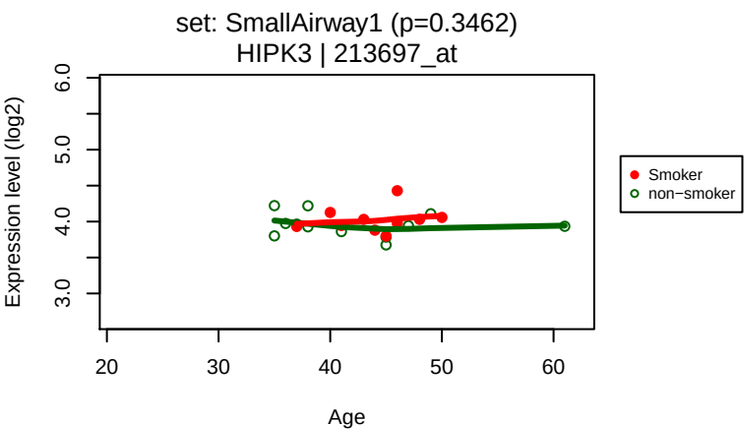
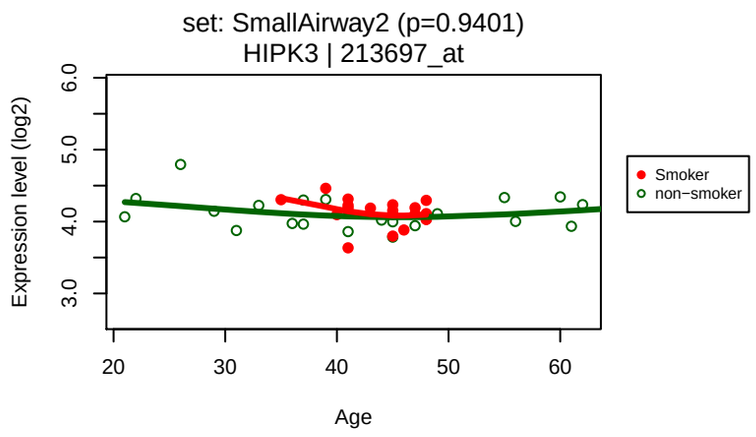
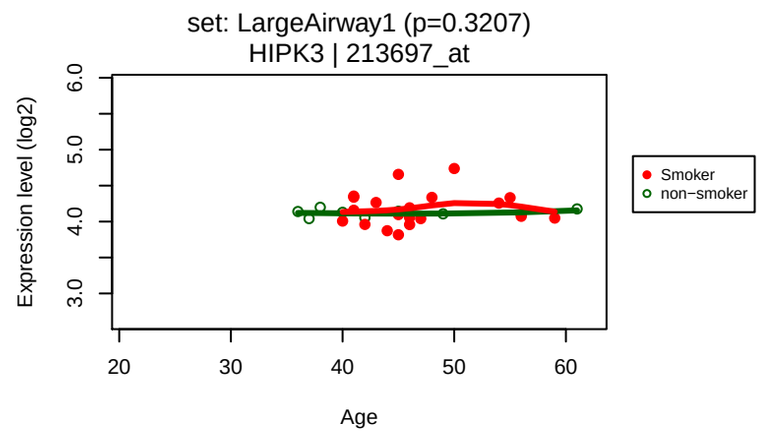

Gene expression of smokers and nonsmokers
gene: HIPK3 | 213697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3462
p=0.9401
p=0.5535
p=0.3207
p=0.033
p=0.097
p=0.0227
p=0.7446
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

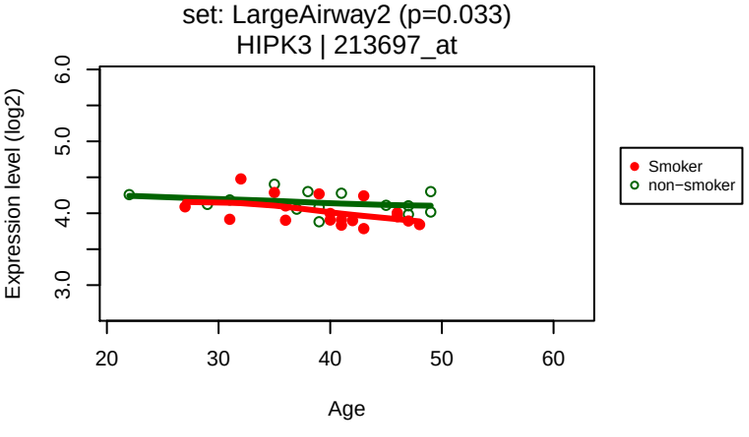

Gene expression ratio (smokers/nonsmokers)
gene: HIPK3 | 213697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3462
p=0.9401
p=0.5535
p=0.3207
p=0.033
p=0.097
p=0.0227
p=0.7446
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

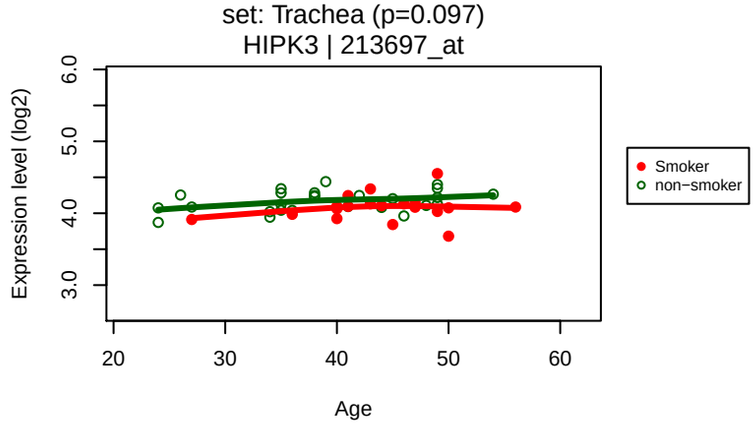

Gene expression ratio in Males
gene: HIPK3 | 213697_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8459
p=0.0287
p=0.0698
p=0.5866
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK3 | 213697_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7167
p=0.7269
p=0.4605
p=0.8534
SmallAirway2
LargeAirway2
Trachea
Macrophage2

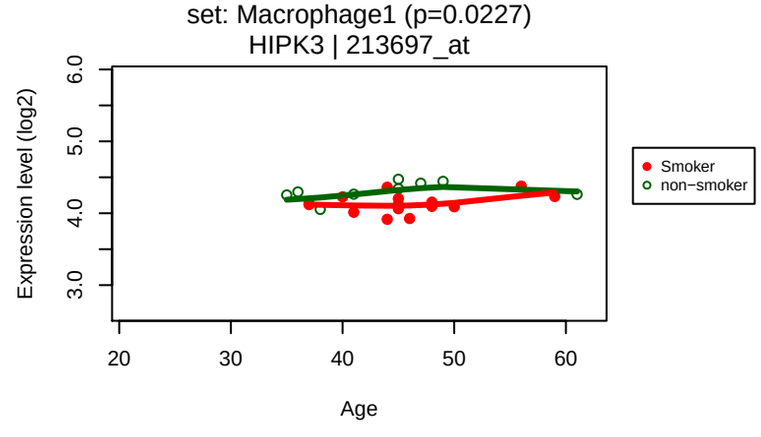

set: SmallAirway1 (p=0.3462)
HIPK3 | 213697_at
Expression level (log2)
Age
Smoker
non−smoker

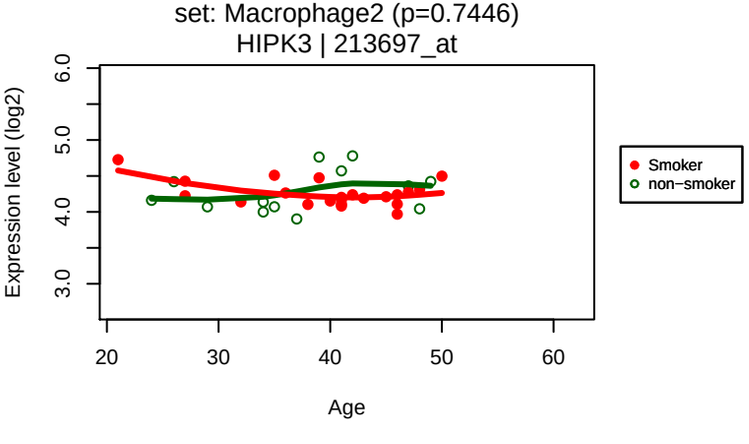

set: SmallAirway2 (p=0.9401)
HIPK3 | 213697_at
Expression level (log2)
Age
Smoker
non−smoker

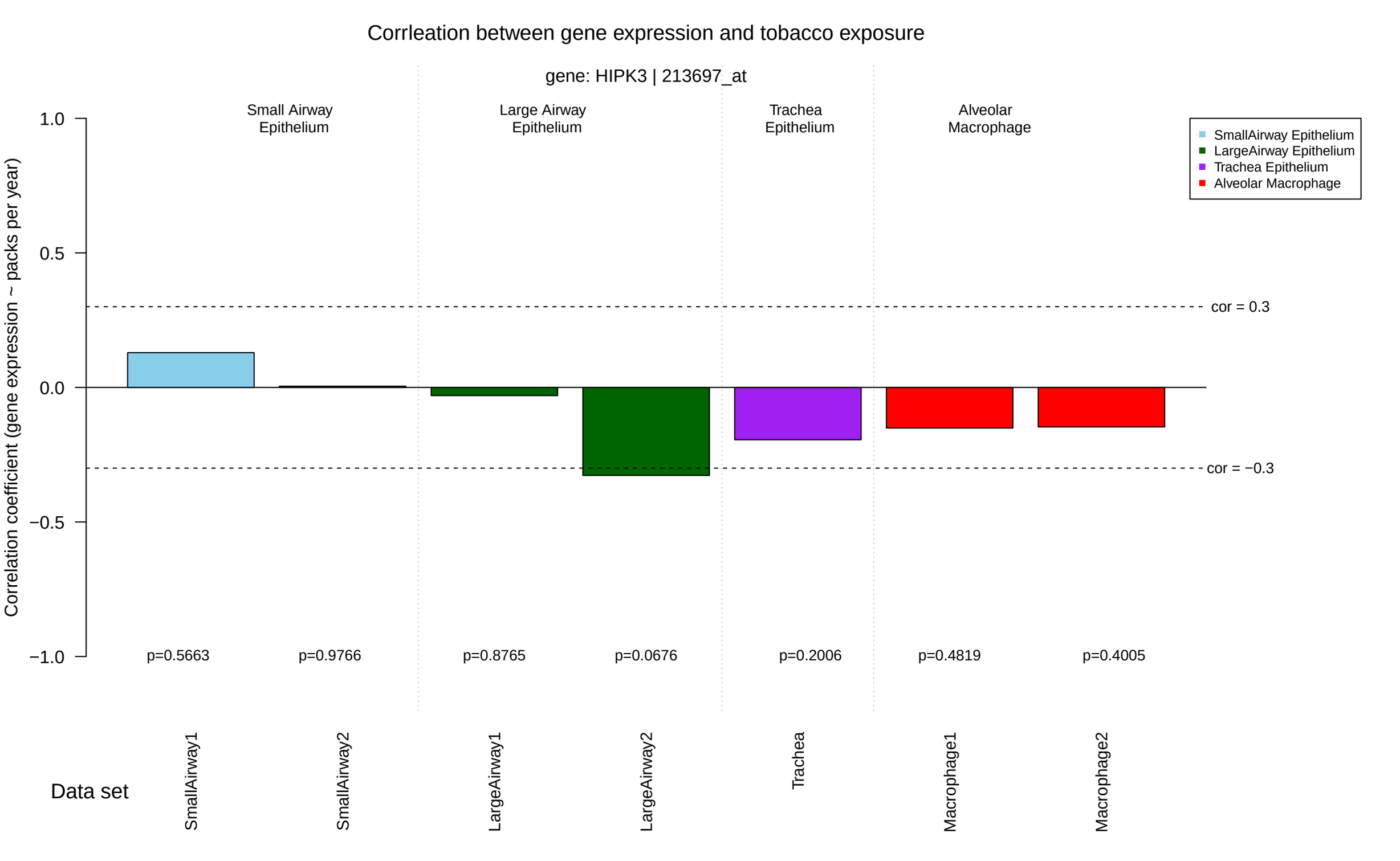

set: LargeAirway1 (p=0.3207)
HIPK3 | 213697_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.033)
HIPK3 | 213697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.097)
HIPK3 | 213697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0227)
HIPK3 | 213697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7446)
HIPK3 | 213697_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK3 | 213697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5663
p=0.9766
p=0.8765
p=0.0676
p=0.2006
p=0.4819
p=0.4005
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2